

Unit tests

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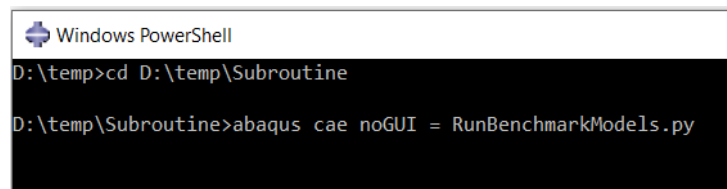
1) Running the script

The verifications can be performed by running the script **RunBenchmarkModels.py**. It can be done by using either ABAQUS Command or User Interface.

a) Method 1: using ABAQUS command

For WINDOW users

- Open **ABAQUS Command**
- To change the directory to the current working directory, type the following command
 - `cd directory path`. For example, `cd D:\temp\Subroutine`
 - Enter
- To run the script, type the following command.
 - `abaqus cae noGUI = RunBenchmarkModels.py`
 - Enter
 - It is noted that “abaqus” can be changed that depends on setting of user’s computer (e.g. `abq614`, `abq2020`, ...)



```
Windows PowerShell
D:\temp>cd D:\temp\Subroutine

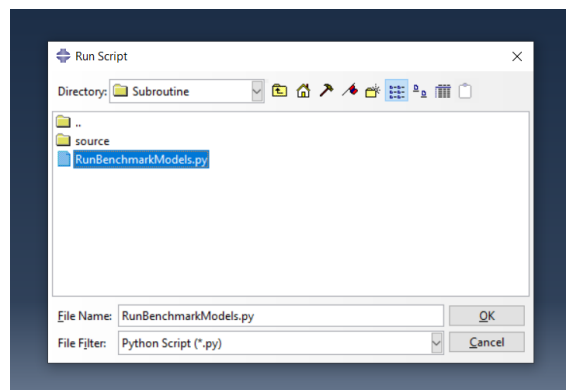
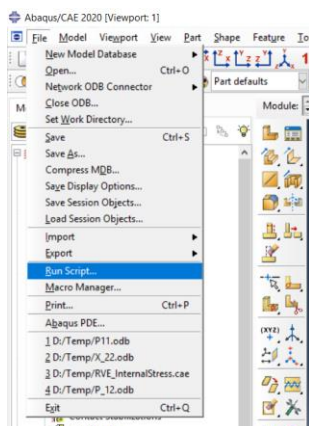
D:\temp\Subroutine>abaqus cae noGUI = RunBenchmarkModels.py
```

For LINUX users

- Open terminal in the current working directory
- To run the script, type the following command in CMD
 - `abaqus cae noGUI = RunBenchmarkModels.py`
 - Enter
 - It is noted that “abaqus” can be changed that depends on the setting of the user’s computer (e.g. `abq614`, `abq2020`, ...)

b) Method 2: using ABAQUS User Interface

- Open **ABAQUS CAE**
- File → Run Script
- Select **RunBenchmarkModels.py**
- OK



2) Introduction and notices

a) Test cases

- There are a total of 8 unit tests corresponding to 8 input files (.inp files)
 - Iso-hardening and monotonic loading
 - Elongation in x-direction (i.e. Ux)
 - Elongation in y-direction (i.e. Uy)
 - Elongation in z-direction (i.e. Uz)
 - Shear deformation γ_{xy} (i.e. ShearXY)
 - Shear deformation γ_{xz} (i.e. ShearXZ)
 - Shear deformation γ_{yz} (i.e. ShearYZ)
 - Superalloy under cyclic load (i.e. SuperAlloy)
 - Austenite with kinematic hardening rule under cyclic load (i.e. Kinematic)

```
# Test cases
Cases = []

## Iso-hardening: Please deactivate the option for kinematic hardening in mod_wkcoup.f (i.e. Iwkcoup_bk = 0)
Cases.append('Ux')
Cases.append('Uy')
Cases.append('Uz')
Cases.append('ShearXY')
Cases.append('ShearXZ')
Cases.append('ShearYZ')

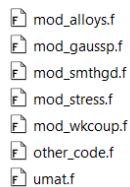
## Superalloy: Please activate the option for SuperAlloy in mod_wkcoup.f (i.e. Iwkcoup_int = 1, Iwkcoup_sup = 1)
Cases.append('SuperAlloy')

## Kinematic hardening: Please active the option for kinematic hardening in mod_wkcoup.f (i.e. Iwkcoup_bk = 1)
#Cases.append('Kinematic')
```

- For the default setting, the script runs test cases for materials with isotropic hardening rules and superalloy.
- Removing or adding symbol “#” to select test cases
- For options of weak coupling effects,
 - Iso-hardening: Please deactivate the option for kinematic hardening in mod_wkcoup.f (i.e. Iwkcoup_bk = 0), otherwise UMAT automatically loads Austenite module
 - Superalloy: Please activate the option for SuperAlloy in mod_wkcoup.f (i.e. Iwkcoup_int = 1, Iwkcoup_sup = 1)
 - Kinematic hardening: Please activate the option for kinematic hardening in mod_wkcoup.f (i.e. Iwkcoup_bk = 1)

b) UMAT files

Folder ‘source’ stores UMAT files.



- mod_alloys.f
- mod_gaussp.f
- mod_smthgd.f
- mod_stress.f
- mod_wkcoup.f
- other_code.f
- umat.f

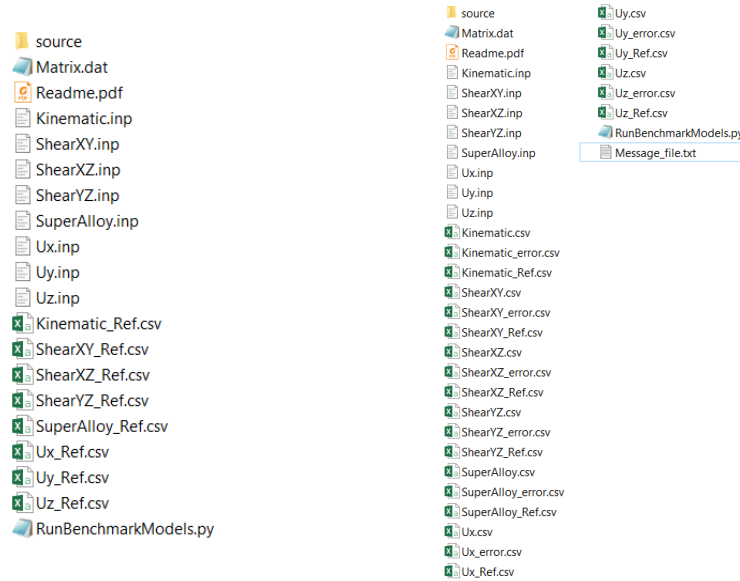
If users want to change the location of UMAT files, ‘sourcePath’ and ‘UmatPath’ variables in the script **RunBenchmarkModels.py** should be updated.

```

13  #=====
14  #
15  executeOnCaeStartup()
16  Mdb()
17
18  sourcePath = os.getcwd() + r'\source'
19  UmatPath = sourcePath + r'\umat.f'
20  PathSave = os.getcwd()
21  Message_file = open('Message_file.txt', 'w')
22

```

c) Necessary files before and after running script



Before running the script

After running the script

Before running the script, the necessary files are listed as follows:

1. Folder “source” stores UMAT files (.f)
2. Matrix.dat should be placed in working directory
3. 8 input files (i.einp) for test cases
4. 8 csv files (i.e. ..._Ref.csv) store reference results

A	B	C	D	E	F
# time (s)	plastic str.	22	33	stress 11 (Mpa)	
0	0	0	0	0	
0.1	0.009314	-0.00465	-0.00465	89.26887	
0.2	0.019128	-0.00955	-0.00955	96.61004	
0.3	0.028847	-0.0144	-0.0144	103.9052	
0.4	0.038475	-0.0192	-0.0192	111.1046	
0.5	0.048011	-0.02396	-0.02396	118.2105	
0.6	0.057457	-0.02867	-0.02867	125.2249	
0.7	0.066817	-0.03334	-0.03334	132.1499	
0.8	0.07609	-0.03796	-0.03796	138.9875	
0.9	0.085279	-0.04254	-0.04254	145.7396	
1	0.094385	-0.04708	-0.04708	152.408	

5. RunBenchmarkModels.py
6. Readme.pdf

After running the script, the additional files are listed as follows

7. 8 csv files (e.g. Ux.csv, Uy.csv, etc.) store results gained from the current simulations

# time (s)	plastic str	22	33] (.)	stress [11] (MPa)
0	0	0	0	0
0.1	0.009314	-0.00465	-0.00465	89.26887
0.2	0.019128	-0.00955	-0.00955	96.61004
0.3	0.028847	-0.0144	-0.0144	103.9052
0.4	0.038475	-0.0192	-0.0192	111.1046
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0.7	0.066817	-0.03334	-0.03334	132.1499
0.8	0.07609	-0.03796	-0.03796	138.9875
0.9	0.085279	-0.04254	-0.04254	145.7396
1	0.094385	-0.04708	-0.04708	152.408

8. 8 csv files (e.g. Ux_error.csv, Uy_error.csv, etc.) store errors of test cases

plastic str	22	33] (.)	stress (MPa)
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.00E+00	0.00E+00	0.00E+00	0.00E+00

9. Message_file.txt shows the maximum errors of each test case

```

Message_file.txt - Notepad
File Edit Format View Help
Maximum_error of case Ux :0.0
Maximum_error of case Uy :0.0
Maximum_error of case Uz :0.0
Maximum_error of case ShearXY :0.0
Maximum_error of case ShearXZ :0.0
Maximum_error of case ShearYZ :0.0
Maximum_error of case SuperAlloy :0.0

```

d) The processes are executed by the script

- Running .inp files
- Output .csv files
- Calculating errors between reference and current results
- Writing errors to .csv files
- Writing maximum errors of each test case to Message.txt
- Deleting unnecessary ABAQUS files such as .odb, .msg, etc.